

THE ABERDARE TECHNICAL COLLEGE

ICT TECHNICIAN LEVEL 5

CAT 2 TERM 3

UNIT: Computer Network Devices and Setup

UNIT CODE: 0612 451 07A

TIME: 1 Hour 30 Minutes

TOTAL MARKS: 50 Marks

NAME: _____

REG NO: _____

DATE: _____

INSTRUCTIONS

- ✓ *Answer ALL questions.*
- ✓ *Answer in a clear and well-structured manner.*
- ✓ *Show all relevant technical knowledge.*

1. Differentiate between a switch and a router in terms of function and OSI layer operation.

(2 Marks)

2. Explain TWO considerations when selecting transmission media for a network installation. (2 Marks)
3. State TWO differences between fibre optic and copper-based network cables. (2 Marks)
4. Describe the role of access points in wireless networking and how they integrate with wired LANs. (2 Marks)
5. Give TWO examples of network security devices and explain their functions in a network. (2 Marks)
6. Differentiate between networking software and network management tools with suitable examples. (2 Marks)
7. Explain TWO functions of a server in a client-server network environment. (2 Marks)
8. State TWO uses of storage systems in a network environment. (2 Marks)
9. Explain the purpose of a multimeter in network installation and troubleshooting. (2 Marks)
10. Differentiate between a patch cord and a backbone network cable in terms of usage. (2 Marks)
11. Describe the function and ONE safety precaution for each of the following tools:
 - a) Cable crimpers (2 Marks)
 - b) Cable strippers (2 Marks)
 - c) Punch down tool (2 Marks)
 - d) Network tester (2 Marks)
 - e) Tone generator and probe (2 Marks)
12. Explain THREE safety precautions when handling network installation tools and equipment. (3 Marks)

13. State TWO applications of fiber optic installation tools in structured cabling. (2 Marks)
14. Explain the role of the following network materials in structured cabling systems:
- a) Network cables (2 Marks)
 - b) RJ45 sockets (2 Marks)
 - c) Patch cords (2 Marks)
 - d) Cable trunking covers (2 Marks)
 - e) Cable ties (2 Marks)
15. State FOUR network components required when designing a Local Area Network (LAN) for a computer laboratory. (4 Marks)
16. Explain TWO considerations when selecting network components for installation. (3 Marks)
17. Describe the step-by-step procedure of setting up a basic wired LAN in a computer laboratory. (4 Marks)
18. Explain TWO challenges that may be encountered during network installation and state how each can be solved. (4 Marks)
19. State TWO tools required during physical network setup and explain their specific uses. (2 Marks)